



Keep Me
Breathing

IMPACT REPORT

*Building
better
futures.*



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Dear Family



To our wonderful new Keep Me Breathing “Family” who have supported us over the past 18 months we would like to offer the biggest thank you for everything you’ve helped us achieve so far, and everything we will achieve in 2025.

Dear Family

A now good friend, and parent of a beautiful CCHS child once told me that having a child with CCHS is akin to planning the family vacation of a lifetime to Thailand ... and then finding out you’ve landed in Italy!

It wasn’t the destination you wanted or planned for. It was missing the tropical beaches, floating markets and exotic wildlife and it feels disappointing...

But then you explore the culture, the food, the history, art and natural beauty and suddenly you realise that you’re experiencing equally amazing things, even if they’re not what you originally planned.

That’s very much what we’ve found with Casper growing up (he is now 3!!!) and always with the relentless pursuit of advanced treatments that will enable all children with CCHS to live with the freedom, independence and health that they deserve. Thank you so much for responding to our calls for help, and allowing us the promise of real change for all children with CCHS.



We owe the thousands of children alive with CCHS today and those born in the future a treatment that can offer freedom, independence, health and a real quality of life

We exist to advance technology and treatments for children (and those who have survived to adulthood) with CCHS, building an innovative solution that will help them lead more independent and fulfilling lives. Parents of children with CCHS from across the world are excitedly waiting for our work to complete, offering a better future for their children which will enable them to thrive and live their best lives.

A special thank you to Eloise and Ash Kent (pictured with us on the left), another Sussex family who have been with us every step of the way. We met the Kents with their son Beau, who also has CCHS, in Brighton Hospital back in 2021 and recognised a similar drive and passion in the search for solutions to make life better for their son. We asked Eloise to come on board as a Trustee of the charity and she has literally worked non-stop. Together they have run events, made incredible connections, and raised a very significant amount of funds for our development programme over this time. We thank them from the bottom of our hearts and can't wait to continue working with them in 2025.

We are a relentless team and will never stop until the job is finished. This report provides you with an update on where we are and where we want to be by the end of the year.

*With all our best wishes
James, Steph, Max & Casper*





2024 in numbers

- Over £500,000 raised in total
- 130+ Doctors, Scientists, Engineers and Medical Professionals have signed an open letter supporting Keep Me Breathing
- KMB Speakers at events like The Beacon Rare Disease Summit and the US CCHS Family Conference
- 20+ events run
- Featured in many local and national news reports including ITV Meridian News & The Times
- 8 corporate sponsors
- £100,000+ awarded in grants
- Over 300 connections made
- Over 1,000 babies born with CCHS in 2024
- Thousands of children with CCHS waiting on us to develop a treatment

Making connections

To build our treatment faster we are constantly looking to connect with others working in the rare disease community, those involved in medical innovation, in healthcare, in value-led corporations or other patient advocacy groups. We attend as many events as we can to network and meet others who can provide support or contribute to our mission. We believe in the mantra, “stronger together” and are always seeking new partnerships or collaborations.

We have forged a close relationship with the incredible team at the CCHS Network in the US.



They have been supporting CCHS families for over 20 years and host a family networking conference every other year, bringing CCHS families together from across the globe. KMB were speakers at the CCHS 2024 Family Conference in California and found it to be an invaluable opportunity to meet patients and researchers in the CCHS arena. Most importantly, to meet and exchange ideas.

Fundraising highlights

A fabulous Leap Year Ball was hosted at Sussex Estate Findon Place to whom we owe a huge thank you. A marquee was donated by Tents N Events and erected in the beautiful grounds. Tickets were sold for a fantastic evening of live music, locally sourced food and drinks as well as a live auction. Generous donors and an array of exciting and interesting prizes such as being a character in the next book written by the blockbusting author, Peter James.



James Howell and Max Edginton cycled over 1,600km from London to Monaco, conquering the French Alps on their way. It took 11 days and there were a few hairy moments up and down rough terrain but they never gave up for us.

We had an incredible 43 people running the 2024 the Brighton Marathon for the charity, some running the 10K and others the whole 26.2 miles. It was a very hot day for April but the whole team finished and made their mark in our eye-catching T-shirts.

**£18,000
raised**



CCHS

Addressing unmet need



Congenital Central Hypoventilation Syndrome or CCHS is a life-long and life-threatening condition and there is no cure.



CCHS causes loss of automatic control of the most basic living function, breathing.



People with CCHS cannot breathe when they sleep and are at risk of sudden death without mechanical ventilation.



80–90% of patients live with a tracheostomy (a surgical opening in their windpipe) to ensure a secure connection to the ventilator.



CCHS is a rare genetic condition with approximately 2,500 diagnosed cases worldwide, and 150 in the UK.



Many experts consider this a significant underestimate as evidence suggests a percentage of SIDS deaths are caused by a mutation in the PHOX2B gene (the gene which causes CCHS). We are working on determining what percentage this is.



Genetic testing for the PHOX2B gene has only been around in recent years, since 2003, and is not widely available.



Ondine's Curse

CCHS was first described as Ondine's Curse. This name comes from a German myth in which a water nymph named Ondine curses her unfaithful husband. According to the legend the curse meant the husband would breathe only when awake and would die if he ever fell asleep.



Making sure the scientific world knows us

James is now a regular speaker at various events. His talk focussed on "Rare People" at the Cambridge Rare Disease Showcase event hosted by Beacon for Rare Disease. Paying tribute to the incredible people who have helped Keep Me Breathing move forward in its work. A special thanks went to Dr Alex Deng, formerly a Clinical Geneticist and now Clinical Lead at the NHS Genomic AI Network. Dr Deng is a Scientific Advisor to the charity and has generously given his time and energy to provide input to shape the direction of our research programme. You can watch some of his videos on our YouTube channel.

Our Innovation Lead, Andy, featured on the left above, has been asked to join the Steering Committee of CCHS Share - an international study which is part-funded by the FDA to look into the long-term needs of people with CCHS.



Spotlight on

I Meet some of the people who have been working incredibly hard for us behind the scenes

ANDREW PACEY, INNOVATION LEAD



“

We have access to better technological tools today than ever before so we are able to harness new ways to innovate which is truly exciting. Our end goal is to develop a treatment that will offer greater independence and improved quality of life for those affected by CCHS.

Our lead engineer Andrew, brings a depth of over 35 years of real-world experience in medical device engineering. He has successfully transitioned medical devices from concept through to market release, demonstrating a strong capability in managing the full lifecycle of highly regulated medical products.

GAYLE HUDSON, PROJECT MANAGER



“

I'm so thrilled to be on board with such incredible people who are single-minded in their work to make a better future for children like Casper and Beau. I am 100% committed to helping them achieve everything we can and more!

Gayle joined the Keep Me Breathing team in March 2024 as the first employee of the charity! With a background in Rare Disease PR and project management she has quickly become integral to KMB as a fast growing med tech charity and we appreciate her tireless determination and work every single day!

ELOISE KENT, TRUSTEE



“

Any spare time I get (night and day) goes into working for KMB. I am full of hope for what the future will hold for Beau once we have advanced treatments available. His life will be truly transformed if we no longer need to use a ventilator or a tracheostomy to help him breathe.

Eloise is Mum to Beau, a beautiful 3 year old boy born with CCHS. Eloise has been on board with KMB almost since day 1. Her background in events has enabled her to devise and run all sorts of fundraisers for the charity which have raised incredible amounts of money to support the charity's work.

Our first book has been published!



“

Dreaming of Breathing...it's beautifully produced with delightful illustrations, suitable for the smallest child and enlightening for the adult reader too.

Dame Jacqueline Wilson

As CCHS is a condition that affects sleep time more than any other time, we decided to bring out a children's bedtime story book. The aim being to raise awareness of CCHS as well as raising funds from its sales.



“Dreaming of Breathing – the adventures of Casper & Beau” is a magical tale based on our two boys, Casper and Beau and follows their adventures in their dreams. It explains how they need help to breathe when they sleep while providing a beautifully written and illustrated bedtime story. We launched it to mark World CCHS Day in 2024 and have sold hundreds of copies already to raise funds for our work.



An advanced technology in 5 years

We have a clear 5 year roadmap to achieve a game-changing treatment following a plan developed by Cambridge Consulting Network, based at Cambridge University. We are developing an implantable biofeedback diaphragm pacer system that will be capable of sensing the patient's ventilatory needs and automatically adjust ventilatory demands to ensure the patient has optimal oxygen and CO2 levels at all times. This device will be implanted and aims to replace the need for mechanical ventilation.

The build for the pacer system consists of three phases



Phase 1: Adaptive Respiratory Modulation Interface

Objective: Develop a high-precision external device that interfaces with our pacer or other pacing systems (i.e. Avery) to dynamically adjust ventilation support based on real-time patient needs via respiratory data from an SP02 monitor (which measures oxygen in the blood). The system will be able to show it can deliver optimised breath timing, volume and frequency, ensuring it matches patient-specific ventilatory requirements.

Due Date: Much of this work has been done and final completion is anticipated in May 2025



An advanced technology in 5 years



Phase 2: Comprehensive Alert and Data Collection System

Objective: Implement an advanced external alert and data review system capable of continuous monitoring and immediate feedback to both caregivers and clinicians.

This system will integrate with cloud-based platforms for remote data review, providing essential alerts and detailed reports on breathing patterns and anomalies. The data collected will allow the interface and control panel to work with AI to control breathing support, therefore reducing the burden on the caregiver.

Due Date: December 2025

Phase 3: Implanted Phrenic Nerve Stimulation Device

Objective: The data from phase 1 and 2 will be used to define the implantable device's requirements. Integration of the hardware systems will allow us to create the final implantable prototype device ready for non-clinical and clinical investigation to assess both safety and efficacy.

Due Date: Mid 2026



Keep Me Breathing is addressing the urgent unmet need for new treatments to improve and save the lives of people diagnosed with CCHS.



Big, Small, Local, National...Our supporters

We are immensely grateful for the support of many local businesses want to take the opportunity to thank them for what they've helped us to achieve.

We are seeking any companies that may be interested in us as their chosen charity for the next 12 months – please get in touch!



Corporate grant of £15,000



Chosen charity for 2025–2026 which will support various initiatives



Sponsorship of events & donations of goods for auctions



Awarded Tesco Stronger Starts Community Grant



Supported various events with corporate sponsorship and provided goods for auction



FINDON
PLACE

Hosted the Spectacular Inaugural Leap Year Ball at their private estate, raising £20,500 for us



Sole sponsor of London To Monaco Bike Ride



Have run multiple charity days for us in local premises

**If you can support us in anyway at all as a company
or individual we would love to hear from you.**

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🌐 www.keepmebreathing.com